

Practice Sheet 1

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These questions are meant for your practice and will not be discussed in tutorials. Feel free to discuss about these in Piazza

1. There are three suspects for a murder: A, B, and C.
 - A says “I did not do it. The victim was an old acquaintance of B’s. But C hated him.”
 - B states “I did not do it. I did not even know the guy. Besides I was out of town all that week.”
 - C says “I did not do it. I saw both A and B downtown with the victim that day; one of them must have done it.”

Assume that the two innocent men are telling the truth, but that the guilty man might not be. Who did it? Deduce the answer by encoding in propositional logic and finding a solution.

2. In an island, there are three tribes : the Knights, Knaves and Normals. The Knights always speak truth, the Knaves always lie, while the Normals lie sometimes and speak truth sometimes. On a visit to this island, I met two inhabitants A and B . A told me that B is a knight and B told me that A is a knave. Prove, using natural deduction, that one of them told the truth but is not a knight, or that one of them told a lie but is not a knave.
3. Let F , G and H be formulas and let $\mathbf{S}$ be a set of formulas. Which of the following statements are true? Justify your answer.
 - (a) If F is unsatisfiable, then $\neg F$ is valid.
 - (b) If $F \rightarrow G$ is satisfiable and F is satisfiable, then G is satisfiable.
 - (c) $P_1 \rightarrow (P_2 \rightarrow (P_3 \rightarrow \dots (P_n \rightarrow P_1) \dots))$ is valid.
 - (d) $\mathbf{S} \models F$ and $\mathbf{S} \models \neg F$ cannot both hold.
 - (e) If $\mathbf{S} \models F \vee G$, $\mathbf{S} \cup \{F\} \models H$ and $\mathbf{S} \cup \{G\} \models H$, then $\mathbf{S} \models H$.

4. Show that a truth assignment α satisfies the wff

$$(\dots (x_1 \leftrightarrow x_2) \leftrightarrow \dots \leftrightarrow x_n)$$

iff $\alpha(x_i) = \text{false}$ for an even number of i 's, $1 \leq i \leq n$.

5. The binary connective **nand**, $F \downarrow G$, is defined by the truth table corresponding to $\neg(F \wedge G)$. Show that **nand** is complete - that is, it can express all binary Boolean connectives.

6. We have seen in class that **HornSAT** has a polynomial satisfiability, while general **SAT** is NP-complete. Here is a reduction called “Hornification” proposed by a student from **SAT** to **HornSAT**. Given a formula φ in CNF, “hornify” each non-horn clause as follows.

- If we have a clause $C_1 = p \vee q \vee r$, then all occurrences of p, q are renamed to $\neg p'$ and $\neg q'$ where p', q' are fresh variables (In general, you could have chosen to rename all but one positive literal to Hornify). Clearly, this renaming can be done in polynomial time.
- Additionally, to respect the relationship between the original variables and their renamed counterparts, add a new Horn clause $p' \wedge p \rightarrow \perp$ (and similarly for q') whenever you rename p as $\neg p'$. This ensures that the new variables p' and q' behave correctly with respect to their original negated forms.

Call the new formula (on an expanded set of variables) as φ' . Since φ' is in **HornSAT**, we can check its satisfiability in polynomial time. Can we conclude that “Hornification” makes **SAT** to be in P?

7. Using natural deduction, show that $\neg y \vdash [x \leftrightarrow ((\neg x \wedge y) \vee (x \wedge \neg y))]$

For the purpose of this question you may use $\leftrightarrow$ introduction rule as follows:

$$\frac{\varphi \rightarrow \psi \quad \psi \rightarrow \varphi}{\varphi \leftrightarrow \psi} \leftrightarrow \text{-introduction}$$

where φ and ψ are wffs.

8. Define a clause of a CNF formula to be *syntactically valid* if it contains both a literal and its complement. Give an efficient algorithm to check the *semantic validity* of a CNF formula and give its time complexity. Argue for the correctness of your algorithm.

A formula F is semantically valid if F evaluates to true in all rows of its truth table.

9. Consider the formula in 3-CNF,

$$\varphi = (x_1 \vee x_2 \vee \neg x_3) \wedge (x_2 \vee x_3 \vee \neg x_4) \wedge (x_1 \vee \neg x_2 \vee x_4)$$

Show that one can construct a graph G_φ from formula φ such that

- G_φ has an independent set of size k iff φ is satisfiable, where k is the number of literals assigned true in the satisfying assignment.
- The size of G_φ is polynomial in the size of φ .
- G_φ must be inspired from φ ; that is, you must have an “encoding scheme” to construct a graph G_φ from any formula φ .

An independent set in a graph is a set of vertices such that no two of them are adjacent.

Give proper explanation to validate that your construction of G_φ is correct.

10. Consider a “black-box” SAT solving algorithm $\mathcal{A}$ that takes as input a propositional formula ϕ and returns whether or not ϕ is satisfiable.

Suppose you have a propositional formula ψ with n variables that you know is satisfiable. Design an algorithm that uses $\mathcal{A}$ as a subroutine to obtain a satisfying assignment for ψ using at most n calls to $\mathcal{A}$.

11. A *renamable Horn formula* is a CNF formula that can be turned into a Horn formula by negating (all occurrences of) some of its variables. For example,

$$(p_1 \vee \neg p_2 \vee \neg p_3) \wedge (p_2 \vee p_3) \wedge (\neg p_1)$$

can be turned into a Horn formula by negating p_1 and p_2 .

Given a CNF-formula F , show how to derive a 2-CNF formula G such that G is satisfiable if and only if F is a renamable Horn formula. Show moreover that one can derive a renaming that turns F into a Horn formula from a satisfying assignment for G .

12. Encode the *eight queens puzzle* as a satisfiability problem. The eight queens puzzle is the problem of placing eight chess queens on the 8×8 chessboard so that no two queens share the same row, column or diagonal.

13. Show that the following statements taken together are inconsistent using Natural Deduction proof system:

S_1 : The following statement is true

S_2 : The previous statement is false

14. Consider a set S of size n . Recall that for a relation over S to be a partial order we require the following to hold:

- (a) Reflexive: $x \preceq x$ for all $x \in S$
- (b) Transitive: $x \preceq y$ and $y \preceq z$ implies $x \preceq z$ for all $x, y, z \in S$
- (c) Antisymmetric: $x \preceq y$ and $y \preceq x$ implies $x = y$ for all $x, y \in S$.

Moreover, $x \in S$ is called a maximal element in $\preceq$ if $x \preceq y$ holds only for $y = x$.

Consider n^2 propositional variables $\{p_{ij}\}_{1 \leq i, j \leq n}$ for an enumeration of $S = (x_1, x_2, \dots, x_n)$ where p_{ij} is set to 1 iff $x_i \preceq x_j$.

Give a formula over these variables which evaluates to $\top$ iff $\preceq$ is a partial order. Also give a formula which evaluates to $\top$ only when $\preceq$ has a maximal element.

15. Define the *length* of a CNF (or DNF) formula as the total number of all literals over all clauses (or all cubes, respectively) in the formula. For example, consider the CNF formula

$$\phi = (x_1 \vee x_2) \wedge (\neg x_1 \vee x_3) \wedge (\neg x_2 \vee x_3 \vee x_4).$$

This formula has length $2 + 2 + 3 = 7$. Similarly, the length of the DNF formula

$$\psi = (x_1 \wedge x_2) \vee (\neg x_1 \wedge x_3 \wedge \neg x_4) \vee (\neg x_2 \wedge x_3 \wedge x_4)$$

is $2 + 3 + 3 = 8$.

Show that there is a family of formulas $F = \{\varphi_n \mid n \in \mathbb{N}\}$ such that:

- Every φ_n is a DNF formula over $O(n)$ propositional variables.
- Every φ_n has length in $O(n)$.
- For every φ_n , there exists no semantically equivalent CNF formula ψ_n (over the same variables as φ_n) such that the length of ψ_n grows polynomially with n .

In order to answer this question, you must:

- (a) Clearly write the DNF formula φ_n ,
- (b) Show that its length is in $O(n)$, and
- (c) Prove that there exists no semantically equivalent CNF formula of length polynomial in n .

16. We know that propositional logic formulas are constructed using symbols in the set $\{\top, \perp, \neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$ in addition to variables, parentheses and commas (if needed). It turns out that such a large set of symbols may not be needed. For example, we have seen in the tutorial that the set $S' = \{\wedge, \neg\}$ suffices to construct a formula that is semantically equivalent to any propositional logic formula.

Let S be a set of symbols that are used in addition to variables, parentheses and commas (if needed) to construct formulas. We say that S is *propositionally expressive* if for every propositional logic formula φ over variables $x_1, \dots, x_n$, there is a semantically equivalent formula over $x_1, \dots, x_n$ that uses only symbols from S in addition to variables, parentheses and commas (if needed). From what we have studied in class and seen in tuts, it should be easy for you to see that $\{\wedge, \neg\}$ is propositionally expressive.

For purposes of this question, we define new ternary logic connectives α , β , γ and δ with the following semantics.

- $\alpha(p, q, r)$ evaluates to 1 (true) iff either both p and q are 1 (true) or p is 0 (false) and r is 1 (true). For example, $\alpha(1, 1, 0) = \alpha(0, 0, 1) = 1$ but $\alpha(1, 0, 1) = \alpha(0, 1, 0) = 0$. You can think of $\alpha(p, q, r)$ as intuitively implementing “if p then q else r ”.
- $\beta(p, q, r)$ evaluates to 1 (true) iff p, q, r all have the same truth value. Thus, $\beta(1, 1, 1) = \beta(0, 0, 0) = 1$, but $\beta(1, 0, 1) = \beta(1, 1, 0) = 0$. You can think of $\beta(p, q, r)$ as intuitively implementing “all of p, q, r are in consensus”.

- $\gamma(p, q, r)$ evaluates to 1 (true) iff exactly one of p, q, r has the value 1 (true). Thus, $\gamma(1, 0, 0) = 1$ and $\gamma(1, 1, 0) = \gamma(0, 0, 0) = 0$.
- $\delta(p, q, r)$ evaluates to the same truth value as the majority of p, q, r . Thus, $\delta(1, 1, 0) = 1$ and $\delta(0, 1, 0) = 0$.

Given below are three sets S of symbols used to construct formulas. In each case you must indicate whether S is propositionally expressive, with justification.

(a) $S = \{\top, \perp, \alpha\}$

(b) $S = \{\top, \gamma\}$

(c) $S = \{\beta, \delta\}$